

AMENDMENT NO. 1 MAY 2026

TO

IS 18718 : 2024 PADDY TRANSPLANTER — TEST CODE AND RECOMMENDATIONS ON SELECTED PERFORMANCE CHARACTERISTICS

(Page 6, clause 8.2.5) — Delete

(Page 6, clause 8.2.6) — Delete

(Page 6, clause 8.2.7) — Delete

(Page 6, clause 8.2.8) — Delete

(Page 7, clause 13.3.1) — Substitute the following for the existing:

‘This test shall be conducted as per IS 12180 (Part 1) during the field performance test of the machine and all the general conditions as specified in 13.1 shall remain the same or under simulated conditions as agreed between the manufacturer and the testing authority.’

[Page 9, clause 14.5.1 (g)] — Delete

(Page 11, clause 14.7.5) — Substitute the following for the existing:

‘14.7.5 Coefficient of Variation, CV

$$CV = \frac{SD}{Mean} \times 100$$

Standard Deviation (SD) is expressed as,

$$SD = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

where

- x_i = value of the i^{th} point in the data set;
- $\bar{x}$ = the mean value of the data set; and
- n = the number of data points in the data.’

(Page 11, clause 14.7.6) — Substitute the following for the existing:

‘14.7.6 Fuel Consumption

$$= \frac{\text{Fuel Consumend(l)}}{\text{Transplanting Area (ha)}},$$

[Page 13, clause 18.1 (b)] — Substitute the following for the existing:

‘Two-hour maximum engine power test under normal ambient condition for diesel engine/rating test for gasoline engine; and’

(Page 13, clause 20) — Insert the following new clauses at the end:

‘21 GUIDELINES FOR ADMINISTRATIVE AND TECHNICAL EXTENSION

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21.1 Administrative Extension

The changes which do not affect paddy transplanter performance such as change in machine colour, decals, design of sheet metal, make and or model for marketing purpose, etc. and which requires physical inspection only, the testing station which carried out the original test can issue an “administrative extension report”. The administrative extension report shall contain a reference to the original test report.

21.2 Technical Extension

21.2.1 The testing station which carried out the original test can issue ‘technical extension report’ by conducting required test(s) on paddy transplanter for certain parameters/improvements such as modification in the cooling system/special characteristics, changes related to statutory/regulatory requirements etc. For reference, a list of parameters eligible for issuing technical extension is given in Table 3. The improvements shall be incorporated permanently/or as an optional feature on paddy transplanter model which had already undergone commercial test. The results shall be reported in the technical extension report. A separate test report shall be released for the technical extension and shall contain the reference of original test report.’

21.2.2 In case the fresh sample is submitted/required for carrying out technical extension, the model will have to be ascertained as being the same model as tested earlier [under initial commercial test (ICT)], by the following checks:

- a) specification as given in the relevant chapter of the relevant ICT Report of the machine;
- b) two hour maximum engine power test under natural ambient conditions for diesel engine/rating test at 100 percent load for a duration of 2 h for gasoline engine; and
- c) nominal speed.

21.2.3 In case of request received for technical extension for certain parameters of the sample, the testing authority may carry out other relevant test(s) also in consultation with the applicant.

21.2.4 If a sample is accepted for technical extension and during test period or subsequently (before release of test report), it is found not being the same model as tested earlier under ICT, the further test on the sample would be stopped and applicant shall withdraw the sample from test and an incomplete test report shall be released.

22 CRITERIA FOR DETERMINING NEW MODEL/VARIANT MODEL

22.1 Base Model

A base model shall be defined as the one, which has defined specifications as reported in the initial commercial test (ICT) report. Manufacturer will specify the base model along with list of variants, which could be added or deleted or supplemented to arrive at variants of the subject base model. The paddy transplanter will be tested as per this standard and initial commercial test report shall be released.

22.1.1 If manufacturer desires, the following features shall be declared as optional features which are not affecting paddy transplanter’s performance with base model, same shall be inspected/verified on the same sample under test and shall be incorporated in initial commercial test (ICT) report. The optional features shall be deemed to be fitted on all the variant(s) of the base model. The optional features components declared on the base model should be submitted along with the ICT paddy transplanter submission. The declaration or fitment of optional features shall not be allowed during the Batch Test.

22.2 Variants

The variant model(s) is determined from the base model by the addition/deletion/supplementing of any one or more of the features given in Table 4. The variants shall be subjected to additional test(s) as stated against each feature of variant.

22.2.1 All the variants which are tested or submitted along with the base model for test shall appear in the test report of the base model.

22.2.2 For variants, the following checks shall be made on a fresh sample only for conformity of sample as being the same as tested earlier:

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- a) specification in full;
- b) nominal speed test; and
- c) two-hour maximum engine power test under natural ambient conditions for diesel engine/rating test at 100 percent load for a duration of 2 hours for gasoline engine.

22.2.3 For variants which can be converted at test site, the variant model can be determined by adding or deleting the variant features in the same sample of base model and if any tests are required, the same can be conducted on the variant machine. If manufacturer desires, a separate variant machine may be submitted, and the relevant tests may be carried out.

22.2.4 Wherever only physical inspection on the variant paddy transplanter is required as per the Table 4, the inspection can be carried out, in consultation with the testing authority.

22.2.5 Beyond the above definition of base model and its variants, other changes would be considered as model change.

22.2.6 Difference between base model, variant and a new model on the basis of engine and other parameters is given in Table 5.'

[Page 14, Table 2, Sl No. (i)(a), col (2)] — Substitute the following for existing:

‘Maximum power under 2 h test, kW for diesel engine’

[Page 14, Table 2, Sl No. (i)(c), col (5)] — Substitute ‘+ 10 percent of declaration (for 2 h max. power for diesel engine & rated power for gasoline engine)’ for ‘+ 5 percent’

[Page 14, Table 2, Sl No. (i)(d), col (5)] — Substitute ‘- 5 percent for diesel engine and not applicable for gasoline engine’ for ‘- 5 percent’

[Page 15, Table 2, Sl No. (i)(g), col (4)] — Substitute ‘Not exceeding 1 percent of SFC at maximum power under high ambient conditions for diesel engine and not applicable for gasoline’ for ‘Not exceeding 1 percent of SFC at maximum power under high ambient conditions’

[Page 15, Table 2, Sl No. (i)(j), col (4)] — Substitute ‘Maximum light absorption coefficient of 3.25/m or Equivalent BOSCH No. 5.2 or 75 Hatridge value for diesel engine and not applicable for gasoline engine’ for ‘Maximum light absorption coefficient of 3.25/m or Equivalent BOSCH No. 5.2 or 75 hatridge value (as per CMVR)’

[Page 15, Table 2, Sl No. (i)(j), col (6)] — Substitute ‘Maximum smoke level shall be reported out of 6 readings tested as per IS 12062 for diesel engine and not applicable for gasoline engines. The observed value should be well within the required limits’ for ‘Maximum smoke level shall be reported out of 6 readings tested as per IS 12062 for diesel engine and same procedure or any other equivalent procedure may be followed for gasoline engines. The observed value should be well within the required limits’

[Page 17, Table 2, Sl No. (viii), col (1) to col (6)] — Substitute the following entries under (a) to (j) for the existing:

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SI No.	Characteristic	Category (Evaluative/Non-Evaluative)	Requirement	Tolerance	Remark
(1)	(2)	(3)	(4)	(5)	(6)
viii)	Discard Limit, mm:				
	a) Cylinder bore	Evaluative			
	b) Piston clearance	Evaluative			
	c) Ring end gap	Evaluative			
	d) Diametrical clearance of big end Bearing	Evaluative			
	e) Axial clearance of big end bearing	Evaluative	Discard limit to be specified by the manufacturer		For such components, values should not exceed their discard limit as specified by the manufacturer
	f) Ring groove clearance	Evaluative			
	g) Thickness of clutch lining	Evaluative			
	h) Diametrical clearance of main bearings	Evaluative			
	j) Crankshaft end float	Evaluative			

[Page 18, Table 2, SI No. (xi)(h) col (2)] — Substitute ‘Size of machine (No. of rows × Spacing between rows, mm)’ for ‘Size of machine’

(Page 18, Table 2) — Insert the following tables at the end:

Table 3 Optional Features of Base Model for issuing Administrative/Technical Extension

(Clause 21.2.1)

SI No.	Optional Features of Base Model	Tests/Inspection that will be Required on Addition of the Features on Base Model
(1)	(2)	(3)
i)	Change in prime-mover make and model within ± 10 percent variation in power and engine speed	No test required. Only physical inspection
ii)	Change in location and actuation system/operating controls like pedal, gear shift levers, system loading/unloading levers etc.	Physical inspection and applicable test, if required
iii)	Change in size of steering wheel/handle and/or change in design of steering system	Physical inspection and turning ability test

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Sl No.	Optional Features of Base Model	Tests/Inspection that will be Required on Addition of the Features on Base Model
(1)	(2)	(3)
iv)	Change in Make/Model of operator's seat (ride on type)	No test required. Only physical inspection
v)	Addition/deletion of gears range modules without changing other speeds	Nominal speed test
vi)	Location of hydraulic pump without change in drive speed	No test required. Only physical inspection
vii)	Location and routing of exhaust system without change in declaration of back pressure	No test required. Only physical inspection
viii)	Change in model/group combination number of FIP/Governor due to addition/deletion of cold starting kit/change in fuel cut-off (on-off) knob/switch without change in engine performance	No test required. Only physical inspection

Table 4 Determination of Variants

(Clauses 22.2 and 22.2.4)

Sl No.	Features of Variant Paddy Transplanter	Additional Tests/Inspection that will be Required on the Feature Variant
(1)	(2)	(3)
i)	Change in prime-mover make/model/engine displacement/rated engine speed/engine aspiration (naturally aspirated/turbo charged), provided declaration in engine power and speed is beyond 10 percent but within 20 percent of base model.	a) Physical inspection; and b) 2 h maximum power test under natural ambient condition for diesel engine/Rating test for gasoline engine.
ii)	Change in fitment/engine accessories such as air compressors, radiators and oil coolers.	a) If the total coolant (water) capacity of radiator is reduced by more than 20 percent that of base model or the oil cooler is removed, then engine test under high ambient condition to be conducted; and b) No additional test required only physical inspection in other cases.
iii)	Addition of number of speeds with add-on modules and/ modifications in gear ratios to meet end customer requirements resulting in variations less than or equal to 20 percent of the nominal speed in maximum speed (forward and reverse) measured on base model.	a) Physical inspection; b) Nominal speed test; and c) Brake test.
iv)	Change in type of FIP — Inline/Rotary/Common rail/CRDI	a) No test, if power declaration is within the tolerance of ± 20 percent of declared values of Base model; and b) If the declaration of power exceeds the tolerance of ± 20 percent of declared values of Base model, engine performance test to be conducted.

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SI No.	Features of Variant Paddy Transplanter	Additional Tests/Inspection that will be Required on the Feature Variant
(1)	(2)	(3)
v)	Change in type of gear box (with or without synchronizers, constant mesh, hydrostatic etc)	a) Physical inspection; and b) Nominal speed test.
vi)	Change in type of drive — wheel drive/4 wheel drive provided the nominal speed remains the same	a) Physical inspection; and b) Turning ability test.
vii)	Fitment of cabin	a) Physical inspection; and b) Noise level test

Table 5 Difference Between Base Model, Variant and a New Model
(Clause 22.2.6)

SI No.	Parameters	Base Model	Variants	New Model
(1)	(2)	(3)	(4)	(5)
i)	Power source	No change	No change	Change
ii)	Principle of operation of Engine (spark/compression ignition) (two stroke/Four Stroke)	No change	No change	Change
iii)	Max. Engine power	No change in declaration	Power declaration exceeding 10 percent but not exceeding 20 percent of the declared values of base model.	Power declaration above 20 percent of declared value of base model
iv)	Make and models of engine	No change	Change subject to the qualifying conditions at SI No. (ii) and (iii) col (4)	Change subject to the qualifying conditions at SI No. (ii) and (iii)
v)	Variation in rated engine rpm	No change in declaration	Change, subject to qualifying condition under SI No. (iv) col (4).	Change, subject to qualifying condition under SI No. (iv) col (5).

NOTES

1 Name of variant model of paddy transplanter must be entirely or partially different from the one of base model.

2 Every modification in the engine parameters to meet the statutory/Regulatory requirements shall be intimated by the manufacturer to the testing authority. The testing authority may consider that the modifications in engine parameters require further tests or otherwise.